Reference-Guided Merge Tree Maps: Quantifying Position–Topology Trade-offs Supplementary Material

Anonymous Authors

This supplement provides derivations, algorithms, data protocols, full metric results, controls, and reproducibility details for the six-sequence, 342-frame evaluation. Standard-metric re-evaluation uses the same fixed configurations and is reported separately from the retained primary anchor-based metrics.

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S1 Formal Derivations and Algorithms

S1.1 Assumptions and feasible layouts

A rooted tree T with leaves $1,\dots,n$; *branching* nodes have ≥ 2 children (unary chains contracted); $p(v)$ is the parent of a non-root branching node v and S_v its leaf set. Join-tree convention: f strictly increases along root-ward paths; leaves are minima (split trees: negate f). Leaf i has reference $q_i \in \mathbb{R}$, width $w_i > 0$, eccentricity radius $h_i = \rho w_i/2$ with $\rho \in [0,1]$; gap $g > 0$; canvas $[a, a+L]$. A layout (π, z, u) satisfies (D) $z_y - z_x \geq (w_x + w_y)/2 + g$ for π-consecutive x, y; (C) $a + w_i/2 \leq z_i \leq a + L - w_i/2$; (E) $|u_i - z_i| \leq h_i$. Assume $\sum_i w_i + (n-1)g \leq L$ so the feasible set is nonempty. $\Pi(T)$: orders in which every S_v is contiguous. $\tau(\pi) = \min_{z,u} \max_i |u_i - q_i|$ subject to (D)(C)(E); $\tau^*(T) = \min_{\pi \in \Pi(T)} \tau(\pi)$; $\tau_{\text{free}} = \min_{\pi \in S_n} \tau(\pi)$.

S1.2 Fixed order

Lemma S1.1 (Anchor elimination)**.** *For fixed z,*

$$\min_u \max_i |u_i - q_i| = \max_i \max(0, |z_i - q_i| - h_i).$$

Hence for $\tau \geq 0$ an admissible u with error $\leq \tau$ exists iff $|z_i - q_i| \leq \tau + h_i$ for all i.

Proof. (E) is separable; $\min_{u \in [z_i - h_i, z_i + h_i]} |u - q_i|$ is the distance from q_i to that interval. □

Lemma S1.2 (Chains with box windows)**.** *Let $[\ell_k, r_k]$, $k = 1..n$, $s_k \geq 0$, $S_{ij} = \sum_{k=i}^{j-1} s_k$. There is z with $z_k \in [\ell_k, r_k]$ and $z_{k+1} - z_k \geq s_k$ iff $\ell_i + S_{ij} \leq r_j$ for all $i \leq j$.*

Proof. ($\Rightarrow$) $z_j \geq z_i + S_{ij} \geq \ell_i + S_{ij}$ and $z_j \leq r_j$. ($\Leftarrow$) $z_j := \max_{i \leq j}(\ell_i + S_{ij})$ satisfies $z_j \geq \ell_j$, $z_j \leq r_j$, and $z_{j+1} \geq z_j + s_j$. □

Proposition S1.3 (Closed form)**.** *Renumber along π; $s_k = (w_k + w_{k+1})/2 + g$, $D_k = \sum_{m<k} w_m + w_k/2 + (k-1)g$, $E_k = \sum_{m>k} w_m + w_k/2 + (n-k)g$. If $\sum_i w_i + (n-1)g > L$ the order is infeasible; otherwise*

$$\begin{aligned}\tau(\pi) = \max\Big\{0,\ &\max_{i<j} \tfrac{q_i - q_j + S_{ij} - h_i - h_j}{2},\\ &\max_k (a + D_k - q_k - h_k),\\ &\max_k (q_k - h_k - a - L + E_k)\Big\}.\end{aligned}$$

Proof. By Lemma S1.1, τ is feasible iff some z satisfies (D)(C) and $z_k \in [\ell_k, r_k]$ with $\ell_k = \max(a + w_k/2, q_k - \tau - h_k)$, $r_k = \min(a + L - w_k/2, q_k + \tau + h_k)$. By Lemma S1.2 this holds iff $\ell_i + S_{ij} \leq r_j$ for all $i \leq j$, i.e. four families of inequalities. (i) $q_i - \tau - h_i + S_{ij} \leq q_j + \tau + h_j$. (ii) $q_i - \tau - h_i + S_{ij} \leq a + L - w_j/2$; since $S_{ij} + w_j/2$ increases in j (increment $w_{j+1} + g$), the binding case is $j = n$ where it equals E_i. (iii) $a + w_i/2 + S_{ij} \leq q_j + \tau + h_j$; since $w_i/2 + S_{ij}$ decreases in i (decrement $w_i + g$), the binding case is $i = 1$ where it equals D_j. (iv) $a + w_i/2 + S_{ij} \leq a + L - w_j/2$, independent of τ; binding at $i = 1, j = n$: capacity. All τ-dependent constraints are lower bounds on τ. □

S1.3 Hierarchy

Lemma S1.4. *$\pi \in \Pi(T)$ iff π arises by permuting, at every node, the blocks of its children and recursing. Thus $|\Pi(T)| = \prod_v \deg(v)!$.*

Proof. Induction on height: the children's leaf sets of the root are contiguous, disjoint, and cover all leaves, so π is a concatenation of these blocks in some order; recurse. □

Proposition S1.5 (Exactness)**.** *$\tau^*(T)$ is attained and is the exact optimum of the model.*

Proof. $\Pi(T)$ is finite and each $\tau(\pi)$ is attained (Prop. S1.3). □

Proposition S1.6 (Monotonicity and decomposition)**.** *Let* $\mathrm{flat}(T,V)$ *delete the non-root internal nodes in* V*, attaching their children to the nearest undeleted ancestor. (i)* $V \subseteq V' \Rightarrow \Pi(\mathrm{flat}(T,V)) \subseteq \Pi(\mathrm{flat}(T,V'))$*, hence* τ^* *is non-increasing. (ii) Deleting all non-root internal nodes gives* τ_{free}*. (iii)* $H := \tau^*(T) - \tau_{\mathrm{free}} \geq 0$*.*

Proof. Surviving nodes keep their leaf sets, so an order making more leaf sets contiguous makes a subset of them contiguous. The star tree only constrains the root. □

Proposition S1.7 (Witness triples)**.** *If* $i, j \in S_v$*,* $k \notin S_v$ *and* $q_i < q_k < q_j$*, then* $\tau^*(T) \geq \frac{1}{2}\min(q_k - q_i,\ q_j - q_k)$*.*

Proof. In any $\pi \in \Pi(T)$, k precedes both i, j or follows both. Anchors are ordered as π: if x precedes y then $u_x \leq z_x + w_x/2 < z_y - w_y/2 \leq u_y$. If k precedes i, $(q_k - u_k) + (u_i - q_i) \geq q_k - q_i$, so one error is $\geq (q_k - q_i)/2$; symmetrically if k follows j. □

S1.4 Exact algorithm at display resolution

Pixels of width λ, canvas N pixels, integer widths W_i, gap G. Leaf i occupies $[s_i, s_i + W_i)$, centre $s_i + W_i/2$; consecutive x, y: $s_y \geq s_x + W_x + G$; $0 \leq s_i \leq N - W_i$. By Lemma S1.1 error $\leq \tau$ iff $s_i \in [lo_i(\tau), hi_i(\tau)]$ with $lo_i = \max(0, \lceil q_i - \tau - h_i - W_i/2 \rceil)$, $hi_i = \min(N - W_i, \lfloor q_i + \tau + h_i - W_i/2 \rfloor)$. For a subtree S and integer x, let $E_S(x)$ be the minimum over legal orders of S and window-feasible placements with all starts $\geq x$ of $s_{\mathrm{last}} + W_{\mathrm{last}} + G$ (∞ if none).

Lemma S1.8 (Earliest placement)**.** *For a fixed order and lower bound* x*, the greedy placement* $s_1 = \max(x, lo_1)$*,* $s_{k+1} = \max(s_k + W_k + G, lo_{k+1})$ *is feasible iff any placement is, and minimises every* s_k*.*

Proof. Induction: any feasible placement has $s_1 \geq \max(x, lo_1)$ and $s_{k+1} \geq s_k + W_k + G \geq s_k^{\mathrm{greedy}} + W_k + G$, $s_{k+1} \geq lo_{k+1}$. If greedy exceeds some hi_k, so does every placement. □

Theorem S1.9 (Exact DP)**.** *For a leaf,* $E_i(x) = \max(x, lo_i) + W_i + G$ *if* $\max(x, lo_i) \leq hi_i$*, else* ∞*. For a node with children* $C_1..C_m$*,*

$$E_S = \min_{\sigma} E_{C_{\sigma(m)}} \circ \cdots \circ E_{C_{\sigma(1)}}.$$

This is computed as $\min(E_B \circ E_A, E_A \circ E_B)$ *for* $m = 2$ *and by subset DP* $F_\emptyset = \mathrm{id}$*,* $F_M = \min_{c \in M} E_c \circ F_{M \setminus c}$ *in general.* τ *is feasible iff* $E_{\mathrm{root}}(0) < \infty$*; feasibility is monotone in* τ*, so bisection yields* τ^*_{disc}*. Cost:* $O(nN)$ *per test for binary trees;* $O(2^m mN)$ *per m-ary node.*

Proof. Each E_S is non-decreasing (a minimum over a feasible set that shrinks as x grows); compositions and pointwise minima preserve this. By Lemma S1.4 a legal order of S is a permutation of child blocks with legal orders inside. For a fixed child permutation, by Lemma S1.8 and monotonicity, the only information a block passes to the next is its earliest next start, and a smaller value never hurts; hence composing the children's optimal E functions is optimal for that permutation, and the minimum over permutations gives E_S (induction on height). Windows widen with τ. □

Proposition S1.10 (Discretisation)**.** *If widths and gap are integer multiples of* λ *and the canvas is grid-aligned, then* $0 \leq \tau^*_{\mathrm{disc}} - \tau^*_{\mathrm{cont}} \leq \lambda/2$*.*

Proof. Discrete layouts are continuous layouts. Conversely round every start of a continuous optimum by $x \mapsto \lfloor x + \frac{1}{2} \rfloor$ (translation-equivariant; not ties-to-even): it is monotone and $\lfloor x + d + \frac{1}{2} \rfloor = \lfloor x + \frac{1}{2} \rfloor + d$ for integer d, so (D) and (C) survive; each centre moves by $\leq \lambda/2$. □

Proposition S1.11 (Certified lower bound)**.** *For arbitrary real widths and gap, let the relaxed discrete problem use* $\lfloor w_i/\lambda \rfloor$*,* $\lfloor g/\lambda \rfloor$ *and the original slacks* h_i*. Then* $\tau^*_{\mathrm{cont}} \geq \tau^*_{\mathrm{disc,relax}} - \lambda/2$*.*

Proof. Every layout feasible for the original problem is feasible for the relaxed continuous problem (smaller widths and gap only loosen (D)(C); (E) is unchanged), so $\tau^*_{\mathrm{relax}} \leq \tau^*_{\mathrm{cont}}$; Proposition S1.10 gives $\tau^*_{\mathrm{disc,relax}} \leq \tau^*_{\mathrm{relax}} + \lambda/2$. □

Remark S1.12. Theorem S1.9 is pseudo-polynomial in the resolution N and exponential in the degree of m-ary nodes; bisection yields the optimum to a tolerance. The scalable DP computes the certificate; choosing a layout inside the budget still enumerates candidate orders.

S1.5 Topological cost and the fundamental trade-off

For any continuous function ψ on the display interval with leaf anchors u_i define $m_\psi(i,j) = \max_{[u_i,u_j]} \psi$ (join tree), $m_T(i,j) = f(\mathrm{lca}_T(i,j))$ and $d_{\mathrm{top}} = \max_{i\neq j} |m_\psi(i,j) - m_T(i,j)|$. For one leaf, set pairwise error and δ^* to zero; empty pair maxima impose no constraint. Use the standard upward root extension when interpreting this metric as a labelled interleaving distance.

Lemma S1.13. *If u_i, u_j are local minima of ψ, the smallest α at which they lie in one component of $\{\psi \le \alpha\}$ is $m_\psi(i,j)$.*

Proof. Components of sublevel sets on a line are intervals. □

Proposition S1.14. *If the minima of ψ are exactly the n labelled leaves with $\psi(u_i) = f(i)$, then d_{top} equals the ℓ_∞ distance between the cophenetic vectors of T and of the merge tree of ψ, which is the labelled interleaving distance (Munch–Stefanou).*

Proof. By Lemma S1.13 $m_\psi(i,j)$ is the height of $\mathrm{lca}(i,j)$ in the merge tree of ψ; diagonal (leaf) entries agree. □

Theorem S1.15 (Topological price of non-contiguity)**.** *If the leaf set S_v of a non-root node v is not contiguous in the anchor order of a 1-D map, then for every ψ: $d_{\mathrm{top}} \ge \gamma_v/2$ with $\gamma_v = f(p(v)) - f(v)$.*

Proof. Pick $i,j \in S_v$, $k \notin S_v$ with u_k between u_i and u_j. Then $[u_i,u_k] \subseteq [u_i,u_j]$ gives $m_\psi(i,k) \le m_\psi(i,j)$; in T, $m_T(i,j) \le f(v)$ and $m_T(i,k) \ge f(p(v))$. Hence $\gamma_v \le m_T(i,k) - m_T(i,j) = [m_T(i,k) - m_\psi(i,k)] + [m_\psi(i,k) - m_\psi(i,j)] + [m_\psi(i,j) - m_T(i,j)] \le 2d_{\mathrm{top}}$. □

Remark S1.16. Theorem S1.15 needs no assumption that anchors are extrema; only Proposition S1.14 (the interleaving interpretation) does. γ_v is an adjacent merge-height gap, not birth–death persistence. The LCA filling is not generally optimal; the optimal filling (Proposition S1.19) is exact.

Remark S1.17 (Tightness). For $T = ((i,j)_v, k)$ with $f(k) \le f(v) + \gamma_v/2$ and anchors ordered i,k,j, choose ψ with $\max_{[u_i,u_k]} \psi = \max_{[u_k,u_j]} \psi = f(v) + \gamma_v/2$; then every pair has error exactly $\gamma_v/2$.

Corollary S1.18 (Lower frontier)**.** *Let $\Phi_T(\delta) = \tau^*(\mathrm{flat}(T, \{v : \gamma_v \le 2\delta\}))$. Every layout satisfying (D)(C)(E), with any filling, has $\max_i |u_i - q_i| \ge \Phi_T(d_{\mathrm{top}})$.*

Proof. Let V be the set of nodes whose leaf sets are non-contiguous. By Theorem S1.15, $V \subseteq W := \{v : \gamma_v \le 2d_{\mathrm{top}}\}$. All nodes outside V are contiguous, so by Lemma S1.4 $\pi \in \Pi(\mathrm{flat}(T,V)) \subseteq \Pi(\mathrm{flat}(T,W))$ (Prop. S1.6). Hence $\max_i |u_i - q_i| \ge \tau(\pi) \ge \Phi_T(d_{\mathrm{top}})$. □

S1.5.1 Optimal filling and the attainable frontier

Fix a leaf order π and anchors $u_{\pi_1} < \cdots < u_{\pi_n}$. A filling is *value-preserving* if it shows the true leaf values at the anchors, $\psi(u_k) = f(k)$ (join convention; negate f for split trees). Write leaves by map position, $M_{ij} = f(\mathrm{lca}_T(i,j))$ for $i < j$, and

$$C_k = \min_{i \le k < j} M_{ij}, \qquad \ell_k = \max(f(k), f(k+1)), \qquad k = 1, \ldots, n-1.$$

Proposition S1.19 (Optimal filling)**.** *Over value-preserving fillings, the infimum is*

$$\inf_\psi d_{\mathrm{top}} = \delta^*(\pi) = \max\Big\{0,\ \max_k (\ell_k - C_k),\ \tfrac{1}{2}\max_{i<j}\big(M_{ij} - \max_{i\le k<j} C_k\big)\Big\}. \tag{S1}$$

It is attained by the barrier filling $b_k = C_k + \delta^*(\pi)$, where b_k is the maximum of ψ on $[u_k, u_{k+1}]$. If every anchor must remain a strict local minimum, barriers with $b_k = \ell_k$ are raised by $\varepsilon > 0$ and $\delta^*(\pi)$ is approached as $\varepsilon \to 0$ (for $T = ((i,j)_1,k)_6$ with $f = 0, 0.2, 5$ and order i,k,j, the infimum 4 is not attained by a strict-minimum filling). $\delta^*(\pi) = 0$ iff $\pi \in \Pi(T)$.

Proof. For any ψ, let b_k be its maximum on $[u_k, u_{k+1}]$. Then $m_\psi(i,j) = \max(f(i),\dots,f(j),b_i,\dots,b_{j-1})$. Since ψ is continuous and u_k, u_{k+1} lie at the ends, $b_k \geq \ell_k$, so $m_\psi(i,j) = \max_{i\leq k<j} b_k$. Hence $d_{\text{top}} \leq \delta$ iff (a) $b_k \leq M_{ij} + \delta$ for all $i \leq k < j$, i.e. $b_k \leq C_k + \delta$, and (b) $\max_{i\leq k<j} b_k \geq M_{ij} - \delta$. Constraint (b) only improves when barriers rise, so δ is feasible iff $b_k = C_k + \delta$ satisfies $b_k \geq \ell_k$ and (b), i.e. iff $\delta \geq \ell_k - C_k$ for all k and $\max_{i\leq k<j} C_k + \delta \geq M_{ij} - \delta$ for all $i < j$. A filling realising any admissible barriers exists (e.g. piecewise linear through $(u_k, f(k))$ and one peak b_k per gap). If $\pi \in \Pi(T)$, the ultrametric property gives $M_{ij} = \max_{i\leq k<j} M_{k,k+1}$ and $C_k = M_{k,k+1} \geq \ell_k$, so $\delta^* = 0$; conversely $\delta^* = 0$ implies $d_{\text{top}} = 0$, and Theorem S1.15 forces every S_v to be contiguous. □

Remark S1.20. The term $\ell_k - C_k$ is new relative to Theorem S1.15: a shallow extremum placed between two leaves that merge deeply raises their 1-D merge level to at least its own value, whatever the filling. The fixed-leaf term can dominate the adjacent-merge-gap bound; it must therefore be included when evaluating a value-preserving filling. If fillings may also change the values shown at the anchors, as the labelled interleaving distance with diagonal entries permits, the term becomes $\frac{1}{2}(\ell_k - C_k)$: for $T = ((i,j)_v, k)$ with $f = 0, 0, 5$, merges at 1 and 6, and order i, k, j, $\delta^* = 4$ for value-preserving fillings but $2.5 = \gamma_v/2$ otherwise.

Theorem S1.21 (Exact position–topology frontier)**.** *For a value-preserving layout with order π, $\max_i |u_i - q_i| \geq \tau(\pi)$ and $d_{\text{top}} \geq \delta^*(\pi)$, and the bounds are attained simultaneously if non-strict extrema are allowed. If every anchor must remain a strict local minimum, the positional bound is attained and the topological bound can be approached arbitrarily closely. Thus the closure of the achievable set has lower envelope*

$$F_T(\delta) = \min\{\tau(\pi) :\ \pi \in S_n,\ \delta^*(\pi) \leq \delta\}, \qquad \Phi_T(\delta) \leq F_T(\delta),$$

and $F_T(0) = \tau^*(T)$, $F_T(\infty) = \tau_{\text{free}}$.

Proof. Positions enter only through $\tau(\pi)$ (Proposition S1.3) and the filling only through π and the anchor values (Proposition S1.19); the optimal positions for π and the barrier filling can be combined because the barrier filling only prescribes values between consecutive anchors, which remain separated by the gap $g > 0$. $\Phi_T \leq F_T$ is Corollary S1.18. □

Remark S1.22. F_T is computed by enumerating S_n ($n \leq 11$ in the exact-frontier study). The DP for Φ_T is practical only when the flattened tree has bounded child degree: flattening a binary tree can create an n-ary root. Neither this DP nor the exact frontier is claimed to scale to arbitrary relaxed hierarchies.

S1.6 Certifying projected distance changes

Let $D_q(t) = |q_i(t) - q_j(t)|$ and $D_u(t) = |u_i(t) - u_j(t)|$ for features present at times t and s. Put $e_i(t) = |u_i(t) - q_i(t)|$.

Proposition S1.23 (Projected distance-change margin)**.** *The difference between the displayed and reference distance changes satisfies*

$$|(D_u(s) - D_u(t)) - (D_q(s) - D_q(t))| \leq e_i(t) + e_j(t) + e_i(s) + e_j(s) =: E_{ij}(t,s).$$

If $|D_u(s) - D_u(t)| > E_{ij}(t,s) + \eta$, the reference change has the same sign and magnitude greater than η. A uniform per-frame positional budget gives $E_{ij}(t,s) \leq 2(B_t + B_s)$.

Proof. The reverse triangle inequality gives $|D_u(t) - D_q(t)| \leq e_i(t) + e_j(t)$. Apply it at both times and then apply the triangle inequality to the difference. □

Remark S1.24. This is an upper-error guarantee about projected coordinates, not two-dimensional distances or human judgements. It uses achieved errors or feasible upper budgets; a small optimal lower certificate alone cannot justify trusting a particular layout.

Proposition S1.25 (Width-free certificate)**.** *Define*

$$\tau_{\text{pt}}(\pi) = \max\{0, \max_{i \text{ before } j \text{ in } \pi} (q_i - q_j)/2\},$$
$$\tau^*_{\text{pt}}(T) = \min_{\pi\in\Pi(T)} \tau_{\text{pt}}(\pi).$$

For every 1-D map whose anchor order lies in $\Pi(T)$, with any widths, gaps, or eccentricity, and every non-decreasing calibration c of the map axis onto the reference axis, $\max_i |c(u_i) - q_i| \geq \tau^*_{\text{pt}}(T)$; moreover $\tau^*_{\text{pt}} \leq \tau^*$.

Proof. If i precedes j and $q_i > q_j$, then $c(u_i) \le c(u_j)$, so $|c(u_i) - q_i| + |c(u_j) - q_j| \ge q_i - q_j$. The bound $\tau^*_{\mathrm{pt}} \le \tau^*$ holds because Proposition S1.3's pair terms with $w = g = h = 0$ are dropped constraints. For distinct anchor coordinates, attainment for a fixed order follows by setting $c(u_k) = \frac{1}{2}(\max_{i \preceq k} q_i + \min_{j \succeq k} q_j)$. □

S1.7 Display certificate pseudocode

Algorithm 1 Certificate τ^* at display resolution (Theorem S1.9).

Require: hierarchy T; pixel widths W_i, gap G, references q_i, radii h_i; N pixels; tolerance ε

function EARLIEST(S, τ) ▷ array over starts $x = 0, \dots, N$
  if S is a leaf i **then**
    $lo_i \leftarrow \max(0, \lceil q_i - \tau - h_i - W_i/2 \rceil)$, $hi_i \leftarrow \min(N - W_i, \lfloor q_i + \tau + h_i - W_i/2 \rfloor)$
    return $x \mapsto \max(x, lo_i) + W_i + G$ if $\max(x, lo_i) \le hi_i$, else ∞
  $E_c \leftarrow$ EARLIEST(c, τ) for every child c of S
  $F_\emptyset \leftarrow$ id; $F_M \leftarrow \min_{c \in M} E_c \circ F_{M \setminus \{c\}}$ for all subsets M of children
  return $F_{\text{all children}}$ ▷ binary: $\min(E_B \circ E_A, E_A \circ E_B)$

$\tau^* \leftarrow$ smallest τ with EARLIEST(root, τ)$(0) < \infty$, by bisection to ε

S1.8 Construction pseudocode

Algorithm 2 RG-MTM construction with optional relaxation.

Require: augmented input trees T_t with arc samples, extrema R_i, supports A_i; feature tracks; mode
  Parameters: conflict threshold θ; eligible merge gap κ; positional slack β

$d \leftarrow$ principal direction of tracked extremum displacements; $q_i \leftarrow \langle R_i, d \rangle$ scaled to the axis
for $t = 1, \dots, K$ **do**
  $\tau^* \leftarrow \min_{\pi \in \Pi(T_t)} \tau(\pi)$; $\tau_{\text{free}} \leftarrow \min_{\pi \in S_n} \tau(\pi)$; $H_t \leftarrow \tau^* - \tau_{\text{free}}$
  $T' \leftarrow T_t$
  if relaxed mode and $H_t > \theta$ **then** ▷ threshold and prune
    $W \leftarrow \{v : v \text{ is non-root internal, } \gamma_v \le \kappa\}$; $b_0 \leftarrow \widehat{\tau}(\mathrm{flat}(T_t, W))$
    for $v \in W$ by decreasing γ_v **do**
      if $\widehat{\tau}(\mathrm{flat}(T_t, W \setminus \{v\})) \le b_0 +$ half a pixel **then**
        $W \leftarrow W \setminus \{v\}$
    $T' \leftarrow \mathrm{flat}(T_t, W)$
  $B_t \leftarrow \tau^*(T') + \beta$ ▷ error budget
  $(\pi, z, u) \leftarrow$ over $\pi \in \Pi(T')$ with $\tau(\pi) \le B_t$, minimize distance stress, reference error, eccentricity, and motion residual w.r.t. $t-1$ subject to (D)(C)(E) and $|u_i - q_i| \le B_t$ ▷ one QP per order
  $(s, e, \alpha) \leftarrow$ discretize intervals and anchors on the fixed canvas
  $v_t \leftarrow \mathrm{FullDomainFill}(T_t, \pi, s, e, \alpha)$ ▷ original augmented tree
  if relaxed mode **then**
    compute $C_k, \delta^*(\pi)$ against T_t; correct barriers (Prop. S1.19)
  report τ^*, H_t, $\max_i |u_i - q_i|$, and measured raster merge error
Assemble $\mathscr{V} = [v_1 \ \cdots \ v_K]$ using shared scalar colours; annotate achieved errors

S2 Data and Construction Protocols

Notation and pipeline interfaces. Table S1 separates source geometry, display geometry, and scalar values. Time indices are omitted only in single-frame derivations. In the filling derivation, i, j denote positions in π; elsewhere they identify leaves. K counts frames, while the native ST-MTM extent parameter is written K_{native}.

Table S1: Notation used throughout the construction.

Symbols	Meaning and role
F_t, Ω, K	Processed scalar field, common spatial domain, and number of frames.
T_t, T_t', ψ_t	Augmented source tree, ordering hierarchy after optional flattening, and displayed scalar profile.
R_i^t, G_i^t, A_i^t	Extremum position, leaf-support centroid, and support area; the first two need not coincide.
$\mathscr{M}_t, S_R, d$	Previous/current leaf correspondences, global motion scatter, and shared unit reference direction.
q_i, u_i, z_i	Reference target, actual extremum anchor, and display-interval centre, in common length units.
w_i, h_i, g, a, L	Interval width, anchor radius, minimum gap, canvas origin, and fixed extent.
$\pi, \Pi(T)$	All-leaf display permutation and the orders that preserve subtree contiguity.
$\tau^*, \tau_{\text{free}}, H$	Best tree-consistent position error, packing-only optimum, and their difference.
θ, β, B_t	Conflict trigger, extra positional slack, and achieved-error upper budget; length units.
γ_v, κ	Adjacent merge gap and eligible-gap threshold; scalar units, not an error guarantee.
M_{ij}, C_k, b_k	Original merge level for display-ranked leaves, spanning-pair gap bound, and displayed barrier.
$\delta^*(\pi), d_{\text{top}}$	Best filling error for an order and measured error of a rendered column; scalar units.
$P, D_{\text{mean}}, D_{\max}$	NROE (normalized reference-order error) and frame-mean/worst-frame NMLE (normalized merge-level error); reported as percentages.
$N, \alpha_i, v_t, \mathscr{V}$	Raster length, integer anchor pixel, one filled column, and $N \times K$ temporal array.

Table S2: Evaluated fields. Leaf counts refer to the processed extrema and merge trees.

Data	Period / construction	Frames	Grid	Leaves	Tree	Smoothing
ERA5	1999-11-17–2000-01-14	118	49×49	1–8	Join	250 km
Fire	Nov.–Dec. 2019	30	64×64	6–11	Split	50.7 km
OLR	2024-06-01–2024-07-30	60	49×49	2–8	Join	500 km
PM2.5	2000-01-01–2000-01-30	30	49×49	1–5	Split	100 km
Ring	Synthetic expansion	40	14×14	1–7	Split	–
Gaussians	Synthetic motion	64	64×64	3–4	Split	–

Six sequences, 342 frames. ERA5 is sampled every 12 hours; fire uses two-day aggregates; OLR and PM2.5 supply one sample per day. Smoothing denotes the fixed Gaussian scale. January PM2.5 is exploratory.

Pressure data. We retain ERA5 mean sea-level pressure for 1999-11-17–2000-01-14, sampled every 12 hours over the North Atlantic and Europe (25–80°N, 40°W–50°E) [4]. A 49×49 equal-area grid and 250 km Gaussian smoothing define join-tree minima. Smoothing is performed on a buffered domain; only extrema of that extended field inside the window are retained, and other cropped basins are filled to their spill levels. These are processed-field minima, not independently validated cyclone centres.

Fire radiative power. The retained Australian fire observations use 10-km source cells from the November–December 2019 sequence [3]. Starting on 2019-11-01, 60 daily observations are summed into 30 two-day frames, whose start dates end on 2019-12-29. Active source cells are accumulated on a 64×64 layout grid, followed by $\log(1 + \text{FRP})$. A weak smooth background removes zero plateaus. The smallest common scale from the declared ladder keeping at most twelve maxima is 2.5 layout cells (50.7169 km); evaluated frames contain six to eleven leaves. Split-tree maxima describe processed fire-intensity structures, rather than independently tracked individual fires.

Radiation fields. Daily NOAA OLR CDR (PSL Ver01Rev02) [6] covers 2024-06-01–2024-07-30 over 15°S–20°N, 60–150°E. Join-tree minima represent low radiation, not validated convective systems. A 49×49 equal-area grid uses five-cell buffers, priority-flood cropping, and no added imputation; a global-extremum fallback is logged. The fixed ladder (100, 150, 200, 250, 350, 500, 700, 1000 km) selects the smallest common scale with at most eight extrema, independently of performance: 500 km. Source hashes and preprocessing trials are recorded.

PM2.5 concentration. The integrated China dataset (V2, 10-km grid) [2] supplies 30 fields, 2000-01-01–2000-01-30, over 28–38°N, 102–114°E. Files carry 10:30 timestamps without an explicit averaging interval. The same buffered grid and eight-extremum rule select 100 km smoothing for split-tree maxima. Values retain $\mu\text{g m}^{-3}$ units. Metadata

record upstream linear-then-nearest filling; we add no imputation. Per-file hashes identify the local snapshot. January was chosen for the main display after the exploratory seasonal comparison; its performance is a selected case, not an independent validation.

Expanding ring. We reproduce the archived TMTM author generator and workspace for the Ring control [3, 5]: 40 frames on the endpoint-inclusive 14×14 grid over $[0,210]^2$, without added smoothing, interpolation or noise. The generator changes a Gaussian radial profile over time. The extracted split tree has one to seven maxima because a sampled ring is not a single isolated extremum. The source commit, generator/workspace hashes and float32 property values are recorded in `experiments/ring/dataset.py`. The main RG-MTM comparison uses the same automatic fixed-direction reference as the real sequences; the analytic ring radius is not silently substituted for that reference.

Prescribed Gaussian motion. The local construction uses a 64×64 grid on $[0,120]^2$ for 64 frames. Gaussian centres at $(25,25)$ and $(85,85)$ are fixed; a third moves from $(115,60)$ toward $(75,35)$ during the first 80% of the sequence and then stops. A connecting ridge controls the hierarchy between the fixed features. The input has three prescribed blobs but three to four extracted maxima, so a prescribed centre is not identical to a retained tree leaf. The construction is inspired by the motivation in ST-MTM [1], not a claim to reproduce its 128×128 input. Current two-step, 2%-diagonal thresholds yield no clear geometric motion events; order and geometry metrics remain evaluable.

Suite boundaries. The six sequences total 342 frames. ERA5-2014 in historical tables is an archived independent season, not an alias of the main ERA5 sequence. Additional PM2.5 months are selection/robustness controls. The 64-frame perpendicular-bisector experiment is a separate synthetic control; none is added to the main total without an explicit protocol change. All methods share processed fields and overlap-based leaf matching, which isolates layout differences within this extraction but does not validate physical identities. For fire, the active 10-km source cells are binned into a square 64×64 field. Consecutive pairs of days are summed before applying $\log(1 + \mathrm{FRP})$. The candidate Gaussian scales are 1, 1.5, 2, 2.5, 3, 4, 5, 6, and 8 cells. The selected 2.5-cell scale corresponds to 50.7 km; the processed field has 6–11 maxima per frame. A broad 16-cell Gaussian background with coefficient 10^{-3} of the field range removes zero plateaus. This background changes the field on which the tree is computed; the reported topology is the topology of that processed field.

OLR and PM2.5 preprocessing uses a spherical Lambert cylindrical equal-area projection, a 49×49 interior grid, and five-cell buffers. The priority-flood boundary operation can change cropped samples, so it is part of the field definition rather than a display-only operation. Source hashes, scale trials, fallback flags, and per-frame modification statistics appear in the protocol JSON files. The main quantitative comparisons share these processed fields exactly. No claim is made about preserving the topology of the unprocessed observations.

S2.1 Synthetic fields and correspondences

Ring uses the archived author generator associated with TMTM [5]. Its 40 frames sample an expanding radial Gaussian on a 14×14 endpoint-inclusive grid over $[0,210]^2$:

$$f_t(x,y) = p_t \exp\left[-\frac{1}{2}\left(\frac{\|(x,y)-c\| - r_t}{s_t}\right)^2\right]. \tag{S2}$$

The declared parameters are $c = (30.4, 40.6)$, $p_t = 1 + 0.3t$, $r_t = 1.94t$, and $s_t = 12.16 + 0.05t$, with $t = 0, \ldots, 39$. The adapter uses their archived float32 property representations. No smoothing, added noise, interpolation, or random seed is used. The adapter records generator/workspace hashes and transposes the source x,y arrays to time,y,x pipeline order. Grid discretization can create several extrema in a radial field, explaining the observed 1–7 leaves.

The Gaussian control contains 64 frames with two fixed centres and one moving centre. The stored construction has fixed centres at $(25,25)$ and $(85,85)$, and a third centre starting at $(115,60)$, moving toward $(75,35)$ before stopping. The field is sampled on 64×64 points over $[0,120]^2$. Each centre contributes a Gaussian of standard deviation 7 with amplitudes 1, 0.95, and 1.05 respectively. A ridge of amplitude 0.5 and width 9 connects the two fixed centres, computed from distance to their line segment. The moving centre is $c_3(t) = (115,60) + [(75,35) - (115,60)]\min(1, t/51.2)$. This construction is defined in `analysis/task_reference.py`; centre metadata are retained in `replicate_gaussians.json`. Extracted leaves, including grid-induced effects, rather than the intended centre count, define the comparison. There are 3–4 leaves per frame and no clear motion events at the default secondary evaluation setting.

All comparison methods share feature IDs, supports, merge trees, and frame correspondences from the common extraction/tracking code. Tracks are computational associations between extrema, not manual annotations. The automatic reference uses the same matched displacements that enter the evaluation. The protocol provides no held-out direction-fitting split; fixed axes assess direction sensitivity without establishing predictive generalization.

S2.2 Layout objective and operating parameters

For candidate order π, let $\mathscr{P}_\pi = \{(\pi_r, \pi_s) : r < s\}$ and use matched leaf pairs $\mathscr{M}_t$ from the shared-reference construction in the main paper. Define

$$\begin{aligned} E_{\text{ref}} &= \frac{1}{n}\sum_i (u_i^t - q_i^t)^2, \\ E_{\text{dist}} &= \frac{1}{\max(1, |\mathscr{P}_\pi|)} \sum_{(i,j)\in\mathscr{P}_\pi} \left[(u_j^t - u_i^t) - \|G_j^t - G_i^t\|_2\right]^2, \\ E_{\text{motion}} &= \frac{1}{\max(1, |\mathscr{M}_t|)} \sum_{(j,i)\in\mathscr{M}_t} \left[(u_i^t - u_j^{t-1}) - (q_i^t - q_j^{t-1})\right]^2, \\ E_{\text{ecc}} &= \frac{1}{n}\sum_i (z_i^t - u_i^t)^2. \end{aligned} \tag{S3}$$

The evaluated solver weights these normalized terms by $4, 1, 0.5, 0.1$, respectively. Matches are unweighted in this construction. The indices j and i explicitly refer to the previous and current leaf, avoiding an assumption that frame-local IDs stay unchanged. All geometry and reference coordinates use the same length units. The quadratic programs obey the interval constraints and $|u_i - q_i| \le \tau^*(T_t') + \beta$. Motion penalizes deviation from the projected reference displacement, not movement itself. Unmatched features have no temporal term, and frames are optimized sequentially rather than jointly.

The global width scale assigns domain area to half the reference extent. The evaluated settings use $g = L/240$, $\rho = 0.5$, $\beta = L/120$, and $\theta = 0.02L$. Relaxation thresholds are 2%, 20%, and infinity of the sequence's processed scalar range. The heuristic score uses a 4096-pixel grid and a fixed half-pixel pruning tolerance; the final budget uses the continuous optimum. These parameters control computation and geometry; no perceptual calibration study establishes them as visibility thresholds.

S2.3 Baselines and metric compatibility

TMTM and ST-MTM preserve tree order; ST-MTM is reimplemented from its paper description [1], not compared through the original authors' executable. Native Hilbert/Morton retain every processed grid sample without refilling, using a minimal power-of-two square with exterior samples discarded. Curve orientation is fixed with x=column and y=row. Native scalar maxima/minima along a sample interval define pairwise merge errors. Optimal barrier filling is applied to RG-MTM relaxation and the packed-sort control. Thus a comparison concerns both ordering and each stated rendering protocol; it does not isolate the effect of ordering alone.

Packed sorting shares interval constraints and secondary objectives but removes hierarchy. Exact projection is a reference control without a rendered scalar topology, and its projected-coordinate error is zero by definition. Fixed-X is a robustness control. Main position comparisons permit monotone calibration and are optimistic order diagnostics; they cannot validate faithful 2-D distances or native pixel coordinates. Stable source-feature order resolves raster ties consistently. The order-only metric then allows a separate monotone coordinate for each tie-broken leaf, so it remains an optimistic lower diagnostic rather than an attained calibration of tied pixels.

S2.4 Exact baseline configuration

The shared forward ST-MTM runner uses temporal mode and Start=0. Except Ring, parameters are uniform pair weights, $r = 0.95$, temporal weight 0.5, size exponent $\alpha = 1$, minimum continuous anchor spacing 0.01, SLSQP tolerance 10^{-11}, and at most 2000 optimizer iterations. Its fixed leaf extent is $K_{\text{native}} = c_{\text{native}} \sum_i A_i(0)$, where c_{native} maps domain area to half the native canvas span. Ring uses the supplied preset: uniform weights, $K_{\text{native}} = 1$, $r = 0.95$,

temporal weight 1.5, $\alpha = 1$, spacing 10^{-3}, and tolerance 10^{-9}. These are our declared comparison settings, not the original preprint's complete real-data configurations.

Nominal ST-MTM lengths are 4096 pixels for ERA5, OLR, and PM2.5; 2048 for fire; 196 for Ring; and 1024 for Gaussians. Global scaling, padding, and rasterization follow the reimplemented skeleton/filling code. TMTM uses temporal overlap optimization and its native augmented sample order. The positional comparison does not use adaptive raster refinement; separate native-merge checks retain nominal-resolution diagnostics. The ST-MTM parameter-grid evidence is an exploratory control, rather than post hoc selection of the best baseline for every reported metric.

Feature correspondences use a maximum intersection-count Hungarian assignment of leaf supports, accepting positive overlaps only. New leaves receive new track identifiers. Tracks are shared across methods. This differs from manually validated physical trajectories, and overlapping time-window events can reuse the same tracks and frames.

S2.5 Shared-reference calculation

For matched extrema, let $\Delta R^t_{ji} = R^t_i - R^{t-1}_j$. The construction accumulates $S_R = \sum_{t=2}^{K} \sum_{(j,i)\in\mathscr{M}_t} \Delta R^t_{ji} (\Delta R^t_{ji})^\top$ and uses a unit principal eigenvector d. Its sign is oriented toward nonnegative total projected motion; a degenerate scatter requires a supplied direction. References are $q^t_i = \langle R^t_i, d\rangle$, with $a = \min_{x\in\Omega}\langle x, d\rangle$ and $L = \max_{x\in\Omega}\langle x, d\rangle - a$. These bounds are shared across frames. Widths satisfy $w^t_i = cA^t_i$, $c|\Omega| = L/2$, and $h^t_i = \rho w^t_i/2$.

S2.6 Metric roles and correspondence to ST-MTM

Our P, τ^*, and H are specific to the supplied reference and display constraints. The scalar comparison d_{top} uses labelled pairwise merge levels, with sequence summaries D (mean normalized framewise maximum) and $D_{\max}$ (sequence maximum). Motion error ME measures the fraction of clear approach/separation events that are reversed or missed at the declared window and thresholds; it is not participant accuracy.

ST-MTM [1] adopts scale-normalized stress (SNS) and trustworthiness (TW), and defines temporal displacement (TD) as a sum of displayed matched-feature displacements. Our stored stress uses the square root of normalized residuals and extremum locations, whereas its stated SNS uses squared residuals and feature centroids. The sequence mean stress cannot simply be squared to obtain mean SNS. Nearest-neighbor agreement is not TW, and our motion error is not TD. Supplement S3 separately recomputes SNS, TW, and canvas-normalized TD on the four fixed main configurations, using the definitions and valid-frame subsets stated there. TD also depends on display scale and the number of matched steps; a low value alone does not establish faithful source motion. Optimization and total times require uniform stage boundaries and repeated timings on common hardware, which the current evidence does not provide.

The main text compares four primary configurations (TMTM, ST-MTM, RG-MTM-P, RG-MTM-R) within each dataset; their numerical table is collected in Supplement S3. Supplemental all-method tables retain native curves, Packed Sort, fixed-axis sensitivity, and exact-projection controls under their declared protocols. Bold minima are specific to each table's comparison population, not universal winners.

Main algorithm interfaces. Main Section 5 describes the complete sequence from shared reference through measured scalar columns and temporal assembly; the corresponding pseudocode is provided in Supplement S1. The display-resolution certificate recurrence is provided in this supplement. Continuous position budgets use legal-order enumeration at the suite's small feature counts; pruning uses rounded widths and gaps on a 4096-pixel grid and a fixed initial score b_0. A grid optimum is not identified with an unrestricted continuous optimum.

S2.7 Full-domain filling and node semantics

The positional model operates on a contracted branching hierarchy, while the scalar renderer retains the augmented input tree and its regular samples. The implementation validates that each input sample belongs to one critical node or one arc, and that samples are monotone along each arc. This input ownership does not imply one-to-one ownership in the display: resampling can repeat or omit samples, paths can reuse arcs, and interpolation can introduce values absent from the input.

Relation to prior constructions. TMTM recursively allocates integer positions to the augmented tree's original samples, placing branching nodes between child blocks [5]. Our initial filling instead follows ST-MTM's leaf-interval and LCA-path construction (its Section 4.2) [1]. This is a borrowed rendering component, distinct from ST-MTM's ordering and layout optimizer. The certificate-guided order and optional minimax barrier correction are separate stages.

Table S3: How the complete augmented input contributes to a RG-MTM column.

Input entity	Display treatment
Every retained leaf	One extremum anchor and one support interval, including leaves outside a witness triple.
Leaf-arc regular sample	Uniformly resampled into the leaf interval; positions alternate outward from the anchor.
Branching merge node	Its LCA value separates adjacent child blocks in a legal order; all labelled leaf-pair levels encode the branching hierarchy.
Internal-arc sample	Resampled along an LCA path in an inter-interval gap, or along an outer root-to-leaf path.
Root and outer context	Border paths fill the remaining canvas. Finite root endpoints and unary tails are not guaranteed by pairwise merge fidelity.

Sampling conventions. Let α_i be an integer anchor and $[s_i, e_i]$ its interval. Place $f(i)$ at α_i and visit available positions $\alpha_i - 1, \alpha_i + 1, \alpha_i - 2, \alpha_i + 2, \ldots$, skipping positions outside the interval. Regular samples are ordered from the leaf toward its parent. Uniform resampling uses equally spaced indices and nearest-index rounding; an empty arc uses linear interpolation within its own endpoint values. Internal paths allocate integer sample counts proportional to arc sizes, distribute remaining pixels by largest fractional remainder, and concatenate samples in path order. Empty arcs receive a unit allocation weight. Deep–deep gaps first divide their non-LCA pixels proportionally between the two path sizes. Sibling gaps use quadratic interpolation toward the LCA, as in ST-MTM.

Algorithm 3 FullDomainFill: initial scalar column from the augmented input.

Require: Augmented input T; all leaves in order π; integer starts s, ends e, anchors α; pixel count N

Initialize $v[0, \ldots, N-1]$ as unassigned
for each leaf i in π **do**
    $v[\alpha_i] \leftarrow f(i)$
    Resample its leaf-arc values into alternating positions in $[s_i, e_i] \setminus \{\alpha_i\}$
for each consecutive pair (i, j) in π **do**
    $v \leftarrow \mathrm{lca}_T(i, j)$; obtain both paths above the already-filled leaf arcs
    if both leaves are direct children of v **then**
        Place $f(v)$ at the gap midpoint; quadratically interpolate both sides
    else if exactly one leaf is a direct child of v **then**
        Place $f(v)$ beside that leaf interval; resample the deeper path into the remainder
    else
        Split non-LCA pixels by path sizes; place $f(v)$ at the split; resample both paths
Fill both outer regions by resampling paths between the root and the outermost leaves
Verify every pixel is assigned and finite; **return** v

Correction uses the original tree. The flattened tree T' defines admissible orders only. Original merge levels M_{ij} and gap bounds C_k always come from T. In the relaxed mode, each open inter-anchor segment is clipped to its target barrier, and its existing maximum is raised to that barrier if needed. Anchors remain fixed. Split-tree processing negates values before filling/correction and negates them back afterwards. A positive clearance above adjacent leaf values keeps extrema strict when the theoretical optimum would produce a plateau. At least one interior pixel per inter-anchor segment is required; raster layouts that collapse the required separation are rejected or rendered at higher resolution.

Executed five-leaf example. the worked filling example uses $T = (((A_1, A_2)_2, B)_4, (C, D)_3)_8$, with all five leaf values zero and two regular samples on each arc. Here $A = \{A_1, A_2\}$ is a subtree, not a single leaf interval. A tree-consistent

order A_1, A_2, B, C, D has barriers $(2, 4, 8, 3)$ and zero pairwise error. For order A_1, A_2, C, B, D, the gap bounds are $C = (2, 4, 3, 3)$ and $\delta^* = 2.5$. Our minimax construction gives barriers $(4.5, 6.5, 5.5, 5.5)$ and measured maximum error 2.5.

Although A_1, A_2 remain consecutive, their merge moves from 2 to 4.5. Flattened ordering constraints, actually non-contiguous subtrees, and numerically changed merge levels are therefore different sets. This construction minimizes the largest leaf-pair error, not the number of modified relations. Protecting selected internal merge levels would define a different constrained filling problem and is not part of the evaluated method.

Tracing white and grey nodes to pixels. With anchors $(10, 30, 50, 70, 90)$ and four interval pixels on either side, white r_1 has value $2/3$ on the A_1 leaf arc and fills pixels $8, 9, 11, 12$. White r_2 has value $8/3$ on the internal A-to-parent arc and fills pixels 35–39 between A_2 and B. Grey S is the A_1–A_2 merge: $f(S) = 2$ occupies midpoint pixel 20. Thus white nodes supply resampled values, while grey nodes supply merge thresholds; neither receives a leaf's geographic anchor optimization. The exported trace verifies these values against actual pixels. Relaxed pixel 20 carries barrier 4.5, marking a changed relation rather than an unchanged source sample.

Verification and scope. Checks cover all 120 orders, ten leaf pairs, and independent gap-witness feasibility, including failure below the optimal positive budget. These synthetic validations add no real-data outcomes.

S3 Additional Experiments and Validation

S3.1 Metric definitions and aggregation

The six-sequence evaluation distinguishes mathematical optima, achieved anchor displacement, measured raster errors and geometric motion outcomes. Each quantity is aggregated over its stated frame, feature or event population.

Normalized Reference-Order Error (NROE, P). For each frame, stable sorting of displayed anchors gives π. The point-model optimum is $\tau_{\mathrm{pt}}(\pi) = \frac{1}{2}\max\{0, \max_{r<s}(q_{\pi_r} - q_{\pi_s})\}$. Its temporal mean divided by the fixed sequence reference extent L is P. This removes widths, gaps, canvas bounds, and anchor radii, permitting a monotone calibration of distinct anchors. Stable tie-breaking assigns separate calibrated values to tied pixels; it is an optimistic diagnostic in that case. τ^*_{pt} minimizes this point cost over tree-consistent orders. Neither P nor τ^*_{pt} equals achieved reference displacement under the full interval model.

Achieved continuous displacement. For each retained feature-frame anchor define $e_i^t = |u_i^t - q_i^t|$. Report

$$E_{\mathrm{pool}} = \frac{\sum_t \sum_{i=1}^{n_t} e_i^t}{L \sum_t n_t}, \qquad E_\infty = \frac{\max_{t,i} e_i^t}{L}.$$

The original record also saves the pooled 95th percentile. Pooling weights frames by their retained feature count; the maximum weights no frame average. The full certificate τ_t^* instead bounds a frame's maximum, so it must not be compared directly with E_{pool}. Values here use continuous RG-MTM-P and RG-MTM-R anchors in reference units, before raster rounding. TMTM/ST-MTM native axes have different units and width models and are not assigned these direct-reference scores without an explicit calibration protocol.

Table S4: Achieved continuous reference displacement (% of L).

	RG-MTM-P		RG-MTM-R, $\kappa = 20\%$	
Data	E_{pool}	E_∞	E_{pool}	E_∞
ERA5	4.13	26.19	2.14	18.46
Fire	3.18	20.48	0.80	6.64
OLR	1.00	17.14	0.57	1.69
PM2.5	2.30	10.24	1.18	5.60
Ring	2.43	11.90	0.71	2.52
Gaussians	4.18	12.11	1.65	8.07

E_{pool} pools all retained feature-frame anchors; E_∞ is the largest absolute continuous anchor error. These use the saved layouts and do not measure a newly computed layout. Raster rounding and scalar correction are separate. The retained baseline record also reports the pooled 95th percentile.

Normalized Merge-Level Error (NMLE, D). $\Delta_F = \max_{t,x} F_t(x) - \min_{t,x} F_t(x)$ is fixed per processed sequence. $D_t = d_{\text{top}}(\psi_t, T_t)/\Delta_F$ measures the maximum pairwise labelled merge discrepancy in frame t. $D_{\text{mean}} = K^{-1}\sum_t D_t$ averages frames; $D_{\max} = \max_t D_t$ is the worst frame. These are displayed as $\text{NMLE}_{\text{mean}}$ and $\text{NMLE}_{\max}$; the symbols and saved numerical keys are unchanged. The fixed-label cophenetic basis follows existing labelled-tree metrics [7]; the experiment-specific normalization does not create a new unlabelled tree distance. One retained leaf gives zero pairwise error. For split trees the profile and original levels are negated before taking maxima. Native-curve barriers are measured on unchanged sample traversals. The revised main table uses `results/filling.json`'s measured optimal-filling summaries, while `results/eval_projection.json`'s `dstar` is the fixed-order closed-form optimum. Their small clearance/raster discrepancies are checked separately rather than silently identifying an optimum with a measurement.

Motion events and inference. A pair of tracks must be present at t and $t+k$; its true distance is the Euclidean distance between retained extrema. A clear event has distance change exceeding $\varepsilon_{2D} = 0.02\,\text{diag}(\Omega)$ in magnitude. Main results fix $k = 2$. Displayed pair distance uses actual anchors and a method-specific threshold $\varepsilon_m = 0.02A_m$, where A_m is that method's full-sequence occupied anchor range. A nonzero opposite sign is a reversal; a change within $[-\varepsilon_m, \varepsilon_m]$ is a miss. ME is $(N_{\text{rev}} + N_{\text{miss}})/N_{\text{clear}}$. The same true events are compared across methods, but the no-change thresholds depend on declared display ranges. If $N_{\text{clear}} = 0$, error rates are undefined; they are not perfect scores. Motion labels are computational geometry, not participant responses. Projected-reference events use a different truth and threshold and must not be subtracted from 2D-truth rates to estimate projection loss.

Paired differences reuse the same events and employ circular moving-block bootstrap over time, with block length $k+2 = 4$ at the primary setting. The retained intervals are exploratory and have no multiplicity correction. Sparse selected PM2.5 events cannot establish a seasonal superiority claim; threshold/window and fixed-direction results remain controls.

Geometry and stability diagnostics. For unordered extremum pairs in a frame, set $a_{ij} = |u_i - u_j|$, $b_{ij} = \|R_i - R_j\|$, and $\alpha = (\sum a_{ij}b_{ij})/(\sum a_{ij}^2)$. The implementation averages

$$S_t = \sqrt{\frac{\sum_{i<j}(\alpha a_{ij} - b_{ij})^2}{\sum_{i<j} b_{ij}^2}}$$

over frames with at least three leaves. NN is the framewise mean agreement of each extremum's nearest neighbour under a and b, then averaged over the same frames. Stable array order resolves distance ties as implemented. Report the number of eligible frames; degenerate cases need explicit treatment if this protocol is extended beyond the present suite.

Flips counts sign changes in display order among pairs matched across consecutive frames, divided by the number of eligible matched pairs. J averages $|u_i^t - u_j^{t-1}|/A_m$ over individual matches. These stability diagnostics do not compare motion with a source target: true crossings or rapid source motion can increase either score. The QP's distance objective uses support centroids; the retained diagnostic S uses extremum locations. These choices answer different questions and are not silently interchanged.

Table S5: Normalized Reference-Order Error (NROE), motion-event error rate (ME), and Normalized Merge-Level Error (NMLE).

		TMTM		ST-MTM		RG-MTM-P (ours)		RG-MTM-R (ours)			
Data	N_{clear}	NROE↓	ME↓	NROE↓	ME↓	NROE↓	ME↓	NROE↓	ME↓	$\text{NMLE}_{\text{mean}}\downarrow$	$\text{NMLE}_{\text{max}}\downarrow$
ERA5	104	19.23	0.471	16.28	0.413	3.77	0.231	1.16	0.183	1.47	26.07
Fire	112	29.30	0.393	24.02	0.357	5.83	0.321	0.34	0.312	3.96	19.56
OLR	56	19.31	0.446	17.21	0.214	0.89	0.054	0.00	0.054	1.10	33.56
PM2.5	7	11.86	0.714	11.30	0.429	1.11	0.429	0.16	0.286	0.86	9.48
Ring	124	7.18	0.645	10.39	0.347	2.66	0.315	0.13	0.129	2.73	12.35
Gaussians	0	25.76	–	25.76	–	5.43	–	1.38	–	3.35	9.98

NROE is P (% of reference extent); NMLE is D (% of processed scalar range). ME is the fraction of reversals plus misses among the shared N_{clear} events, displayed in $[0,1]$ to three decimals; – means no clear events. The motion protocol uses two steps and 2% thresholds. RG-MTM-R uses $\kappa/\Delta_F = 0.20$, an eligibility threshold rather than an error cap. NMLE uses measured filled-raster barriers; the other three methods have zero measured pairwise merge error.

ST-MTM's stated SNS has squared residuals and centroid distances, TW penalizes introduced neighbours by original-space ranks, and TD sums matched displacements. SNS and TW are adopted from prior work [8, 9], while ST-MTM specifies TD for its matched display positions. NN is not TW, J is not TD, and $(\text{mean}\, S_t)^2$ is not mean (S_t^2). The separate standard-metric re-evaluation below uses the required entities and aggregation without changing these historical diagnostics.

Source mapping. Order statistics use `results/final_performance.json`; motion and geometry use `results/eval_projection.json`; measured barrier costs use `results/filling.json`; achieved anchors use `results/replicate_<dataset>.json`'s `tradeoff` entries at caps 0 and 0.2, normalized by its `certificate.axis`. The main-paper ledger stores source hashes and unrounded values. No metric selection or bold-face ranking is inferred from a combined score.

Method-name mapping. RG-MTM means Reference-Guided Merge Tree Maps. RG-MTM-P is the hierarchy-preserving mode; RG-MTM-R is the relaxed mode. In the saved records, `A` denotes RG-MTM-P and `R 20%` denotes RG-MTM-R at $\kappa = 0.2\Delta_F$. Keys `R 2%` and `R inf` identify the other eligibility settings. Source and result paths use the repository directory `CD-MTM`.

Table S6: Full interval-model certificates: temporal means normalized by reference extent.

Data	Frames	Leaves	Packing %	Hierarchy H %	Total τ^* %	Conflict frames
ERA5	118	1–8	0.16	4.17	4.33	40/118
Fire	30	6–11	0.06	6.12	6.18	19/30
OLR	60	2–8	0.02	0.94	0.96	7/60
PM2.5	30	1–5	0.09	1.31	1.39	5/30
Ring	40	1–7	0.05	2.83	2.88	21/40
Gaussians	64	3–4	0.03	5.95	5.98	47/64

Total is $\overline{\tau^*}/L = \overline{\tau_{\text{free}}}/L + \overline{H}/L$. Conflicts use $H > 0.02L$. These are theoretical diagnostics of each input sequence, not competing method scores; no best-result ranking applies. Values are rounded independently.

Table S7: Motion-proxy error rates (%, ↓) with all seven main methods and RG-MTM-R's worst-frame scalar cost.

Data	Events	TMTM	ST-MTM	Hilbert	Morton	Packed	**RG-MTM-P (ours)**	**RG-MTM-R (ours)**	RG-MTM-R $D_{\max}$
ERA5	104	47.1	41.3	46.2	44.2	**14.4**	23.1	18.3	26.1
Fire	112	39.3	35.7	50.0	48.2	33.0	32.1	**31.2**	19.6
OLR	56	44.6	21.4	48.2	46.4	**5.4**	**5.4**	**5.4**	33.6
PM2.5	7	71.4	42.9	42.9	**28.6**	42.9	42.9	**28.6**	9.5
Ring	124	64.5	34.7	38.7	25.0	**12.1**	31.5	12.9	12.3
Gaussians	0	–	–	–	–	–	–	–	10.0

This complete seven-method comparison complements the primary-method table in the main paper. Bold rates are row minima across the seven shown methods, with all ties; they are numerical minima, not significance claims. Events: common clear-event count at $k = 2$, 2% thresholds. Error = reversal + miss. RG-MTM-R $D_{\max}$ (%, ↓) is the relaxed methods sequence-maximum merge error, not a motion score or cross-method ranking. RG-MTM-P and RG-MTM-R are our two modes. Gaussians has no clear events, so its rates are undefined. Sparse PM2.5 results are descriptive.

Table S8: Expanded numerical comparison. RG-MTM-P and RG-MTM-R are our tree-preserving/relaxed modes. Bold values are minima within each dataset among rendered methods, including Fixed X; ties use unrounded records. Exact projection is a diagnostic control and is excluded from ranking because it has no scalar topology. Underlined P is best among zero-mean-D rows. Stress is Kruskal stress-1; error is reversal plus miss. Ranking populations differ from the main tables, which exclude fixed-axis sensitivity controls.

Data	Method	$P \downarrow$ %	Mean $D \downarrow$ %	Max $D \downarrow$ %	Stress ↓	Events	Error ↓ %
ERA5	TMTM	19.23	**0.00**	**0.00**	0.317	104	47.1
ERA5	ST-MTM	16.28	**0.00**	**0.00**	**0.194**	104	41.3
ERA5	Hilbert	6.75	22.76	49.41	0.326	104	46.2
ERA5	Morton	16.42	25.22	63.63	0.352	104	44.2
ERA5	**RG-MTM-P (ours)**	3.77	**0.00**	**0.00**	0.327	104	23.1
ERA5	**RG-MTM-R (ours)**	1.16	1.47	26.07	0.308	104	18.3
ERA5	Packed sort	**0.00**	6.18	58.66	0.299	104	**14.4**
ERA5	Fixed X (control)	<u>3.77</u>	**0.00**	**0.00**	0.321	104	24.0
ERA5	Exact projection	0.00	–	–	0.310	104	13.5
Fire	TMTM	29.30	**0.00**	**0.00**	0.458	112	39.3
Fire	ST-MTM	24.02	**0.00**	**0.00**	0.256	112	35.7
Fire	Hilbert	29.22	32.98	78.00	0.479	112	50.0
Fire	Morton	31.51	33.72	81.37	0.308	112	48.2
Fire	**RG-MTM-P (ours)**	<u>5.83</u>	**0.00**	**0.00**	0.261	112	32.1
Fire	**RG-MTM-R (ours)**	0.34	3.96	19.56	0.221	112	**31.2**
Fire	Packed sort	**0.00**	7.66	52.58	**0.219**	112	33.0
Fire	Fixed X (control)	31.22	**0.00**	**0.00**	0.397	112	36.6
Fire	Exact projection	0.00	–	–	0.227	112	26.8
OLR	TMTM	19.31	**0.00**	**0.00**	0.379	56	44.6
OLR	ST-MTM	17.21	**0.00**	**0.00**	0.097	56	21.4
OLR	Hilbert	32.50	41.13	78.60	0.358	56	48.2
OLR	Morton	30.76	47.35	76.95	0.528	56	46.4
OLR	**RG-MTM-P (ours)**	<u>0.89</u>	**0.00**	**0.00**	0.084	56	5.4
OLR	**RG-MTM-R (ours)**	**0.00**	1.10	33.56	0.074	56	5.4
OLR	Packed sort	**0.00**	1.10	33.56	0.074	56	5.4
OLR	Fixed X (control)	32.83	**0.00**	**0.00**	**0.074**	56	**3.6**
OLR	Exact projection	0.00	–	–	0.081	56	7.1
PM2.5	TMTM	11.86	**0.00**	**0.00**	0.241	7	71.4
PM2.5	ST-MTM	11.30	**0.00**	**0.00**	**0.137**	7	42.9
PM2.5	Hilbert	0.20	21.84	46.80	0.347	7	42.9
PM2.5	Morton	0.63	21.29	46.80	0.409	7	**28.6**
PM2.5	**RG-MTM-P (ours)**	<u>1.11</u>	**0.00**	**0.00**	0.508	7	42.9
PM2.5	**RG-MTM-R (ours)**	0.16	0.86	9.48	0.443	7	**28.6**
PM2.5	Packed sort	**0.00**	1.78	17.27	0.436	7	42.9

Data	Method	$P \downarrow$ %	Mean $D \downarrow$ %	Max $D \downarrow$ %	Stress $\downarrow$	Events	Error $\downarrow$ %
PM2.5	Fixed X (control)	7.71	**0.00**	**0.00**	0.323	7	**28.6**
PM2.5	Exact projection	0.00	–	–	0.464	7	57.1
Ring	TMTM	7.18	**0.00**	**0.00**	0.451	124	64.5
Ring	ST-MTM	10.39	**0.00**	**0.00**	**0.316**	124	34.7
Ring	Hilbert	11.25	44.55	97.40	0.336	124	38.7
Ring	Morton	11.25	43.32	90.56	0.374	124	25.0
Ring	**RG-MTM-P (ours)**	2.66	**0.00**	**0.00**	0.402	124	31.5
Ring	**RG-MTM-R (ours)**	0.13	2.73	12.35	0.372	124	12.9
Ring	Packed sort	**0.00**	3.44	12.35	0.374	124	**12.1**
Ring	Fixed X (control)	10.87	**0.00**	**0.00**	0.442	124	37.1
Ring	Exact projection	0.00	–	–	0.389	124	13.7
Gaussians	TMTM	25.76	**0.00**	**0.00**	0.584	0	–
Gaussians	ST-MTM	25.76	**0.00**	**0.00**	0.291	0	–
Gaussians	Hilbert	25.76	33.37	33.38	0.340	0	–
Gaussians	Morton	25.76	33.37	33.38	0.242	0	–
Gaussians	**RG-MTM-P (ours)**	5.43	**0.00**	**0.00**	0.343	0	–
Gaussians	**RG-MTM-R (ours)**	1.38	3.35	9.98	0.238	0	–
Gaussians	Packed sort	**0.00**	8.09	12.74	**0.194**	0	–
Gaussians	Fixed X (control)	25.76	**0.00**	**0.00**	0.308	0	–
Gaussians	Exact projection	0.00	–	–	0.216	0	–

Table S9: Retained exact-frontier checks. Half a pixel defines tightness. Selected-order optima exclude secondary-layout slack. These are solver/frontier validation diagnostics, not rankings between datasets.

Data	Max leaves	Tight breaks	Max $F - \Phi \downarrow$ %	RG-MTM-R (ours) on $F \uparrow$	Max excess $\downarrow$ %
ERA5	8	174/290	21.86	118/118	0.00
Fire	11	72/207	11.03	28/30	0.55
OLR	8	175/204	16.31	60/60	0.00
PM2.5	5	29/48	9.29	30/30	0.00
Ring	7	38/133	5.84	39/40	0.14
Gaussians	4	53/147	2.78	64/64	0.00

Table S10: Recomputed scalar filling checks. Bold scalar costs identify the smaller of LCA and optimal filling for the same dataset and setting (ties included). The residual includes strict-extremum clearance and is normalized as recorded.

Data	κ/range	LCA max $\downarrow$ %	Optimal max $\downarrow$ %	Formula residual $\downarrow$
ERA5	0.0	**0.00**	**0.00**	0
ERA5	0.02	1.72	**1.15**	1.1e-05
ERA5	0.2	29.56	**26.07**	1.1e-05
ERA5	∞	64.50	**58.66**	1.1e-05
Fire	0.0	**0.00**	**0.00**	0
Fire	0.02	1.90	**1.66**	2.8e-07
Fire	0.2	19.65	**19.56**	3.4e-07
Fire	∞	52.92	**52.58**	3.4e-07
OLR	0.0	**0.00**	**0.00**	0
OLR	0.02	1.85	**1.79**	1.3e-06
OLR	0.2	33.59	**33.56**	1.5e-06
OLR	∞	33.59	**33.56**	1.5e-06
PM2.5	0.0	**0.00**	**0.00**	0

Data	κ/range	LCA max ↓ %	Optimal max ↓ %	Formula residual ↓
PM2.5	0.02	0.00	**0.00**	0
PM2.5	0.2	10.37	**9.48**	6.1e-07
PM2.5	∞	25.80	**17.27**	6.1e-07
Ring	0.0	**0.00**	**0.00**	0
Ring	0.02	4.32	**2.16**	0
Ring	0.2	24.69	**12.35**	0
Ring	∞	24.69	**12.35**	0
Gaussians	0.0	**0.00**	**0.00**	0
Gaussians	0.02	**0.00**	**0.00**	0
Gaussians	0.2	19.97	**9.98**	1.9e-17
Gaussians	∞	24.84	**12.42**	1.9e-17

Table S11: Retained exploratory PM2.5 seasonal audit: motion error percentages. January was selected for display after this comparison. Bold rates are minima among the three shown methods within each month, with ties; event counts are small.

Month	Frames	Conflicts	Events	ST-MTM ↓	**RG-MTM-P (ours)** ↓	**RG-MTM-R (ours)** ↓
Jan	30	5	7	42.9	42.9	**28.6**
Apr	30	3	1	**0.0**	**0.0**	**0.0**
Jul	30	4	8	12.5	**0.0**	12.5
Oct	30	0	3	100.0	**0.0**	**0.0**

Table S12: Retained direction sensitivity: conflict shares at 2% across eight fixed directions.

Data	Minimum conflict %	Maximum conflict %
ERA5	11.0	34.7
Fire	50.0	76.7
OLR	5.0	50.0
PM2.5	0.0	30.0

Table S13: Complete spatial and temporal diagnostics for the six-dataset suite.

Data	Method	Flips↓	Jitter↓	Stress↓	NN↑
ERA5	TMTM	0.092	0.046	0.317	0.746
ERA5	ST-MTM	0.050	**0.027**	**0.194**	**0.877**
ERA5	Hilbert	0.078	0.059	0.326	0.726
ERA5	Morton	**0.014**	0.046	0.352	0.679
ERA5	Packed sort	0.078	0.086	0.299	0.672
ERA5	**RG-MTM-P (ours)**	0.110	0.101	0.327	0.731
ERA5	**RG-MTM-R,** 20% **(ours)**	0.073	0.089	0.308	0.700
ERA5	**RG-MTM-R,** ∞ **(ours)**	0.083	0.086	0.296	0.682
ERA5	Exact projection	0.078	0.086	0.310	0.660
Fire	TMTM	0.334	0.102	0.458	0.639
Fire	ST-MTM	0.076	0.053	0.256	**0.753**
Fire	Hilbert	0.016	0.020	0.479	0.559
Fire	Morton	0.010	**0.015**	0.308	0.530
Fire	Packed sort	**0.005**	0.018	0.219	0.594
Fire	**RG-MTM-P (ours)**	0.094	0.054	0.261	0.658

Data	Method	Flips↓	Jitter↓	Stress↓	NN↑
Fire	**RG-MTM-R,** 20% **(ours)**	0.026	0.022	0.221	0.630
Fire	**RG-MTM-R,** ∞ **(ours)**	0.010	0.019	**0.219**	0.600
Fire	Exact projection	0.005	0.017	0.227	0.589
OLR	TMTM	0.181	0.089	0.379	0.708
OLR	ST-MTM	0.007	**0.016**	0.097	0.913
OLR	Hilbert	0.051	0.078	0.358	0.691
OLR	Morton	0.130	0.085	0.528	0.508
OLR	Packed sort	**0.000**	0.058	**0.074**	**0.915**
OLR	**RG-MTM-P (ours)**	**0.000**	0.061	0.084	0.912
OLR	**RG-MTM-R,** 20% **(ours)**	**0.000**	0.058	**0.074**	**0.915**
OLR	**RG-MTM-R,** ∞ **(ours)**	**0.000**	0.058	**0.074**	**0.915**
OLR	Exact projection	0.000	0.061	0.081	0.918
PM2.5	TMTM	0.048	0.081	0.241	0.771
PM2.5	ST-MTM	**0.000**	**0.027**	**0.137**	**0.903**
PM2.5	Hilbert	0.190	0.191	0.347	0.515
PM2.5	Morton	0.048	0.194	0.409	0.469
PM2.5	Packed sort	0.095	0.189	0.436	0.547
PM2.5	**RG-MTM-P (ours)**	**0.000**	0.201	0.508	0.588
PM2.5	**RG-MTM-R,** 20% **(ours)**	0.143	0.198	0.443	0.601
PM2.5	**RG-MTM-R,** ∞ **(ours)**	0.095	0.190	0.433	0.581
PM2.5	Exact projection	0.095	0.184	0.464	0.568
Ring	TMTM	0.118	0.034	0.451	0.429
Ring	ST-MTM	0.030	0.030	**0.316**	**0.764**
Ring	Hilbert	**0.018**	**0.021**	0.336	0.677
Ring	Morton	**0.018**	0.023	0.374	0.657
Ring	Packed sort	0.022	0.031	0.374	0.539
Ring	**RG-MTM-P (ours)**	0.114	0.054	0.402	0.494
Ring	**RG-MTM-R,** 20% **(ours)**	0.059	0.037	0.372	0.553
Ring	**RG-MTM-R,** ∞ **(ours)**	0.059	0.037	0.372	0.553
Ring	Exact projection	0.022	0.029	0.389	0.560
Gaussians	TMTM	0.004	0.003	0.584	0.445
Gaussians	ST-MTM	**0.000**	0.002	0.291	0.605
Gaussians	Hilbert	**0.000**	**0.001**	0.340	0.605
Gaussians	Morton	0.004	0.003	0.242	0.801
Gaussians	Packed sort	0.004	0.003	**0.194**	**0.926**
Gaussians	**RG-MTM-P (ours)**	**0.000**	0.003	0.343	0.605
Gaussians	**RG-MTM-R,** 20% **(ours)**	0.004	0.005	0.238	0.790
Gaussians	**RG-MTM-R,** ∞ **(ours)**	0.004	0.004	0.195	0.900
Gaussians	Exact projection	0.004	0.003	0.216	0.926

Flips and jitter describe stability, not agreement with physical motion. Stress and NN use original extremum distances on frames with at least three extrema. Bold entries mark per-dataset numerical optima among displayed rendered methods, including ties; exact projection is a diagnostic control excluded from ranking. The main text discusses selected comparisons from this complete control population.

S3.2 Standard-metric re-evaluation

The primary NROE and NMLE measures answer the reference-order and labelled-merge questions. To test their relationship to established criteria, we separately reconstructed the same six datasets and four fixed main configurations and saved their actual raster anchors. No parameter search or change to the scientific algorithms was performed. The resulting standard metrics are additional measurements, rather than renamed historical *S*/NN/*J* values.

Geometry and stability. SNS follows the scale-normalized squared residual used by ST-MTM [1,8], using pairwise support-centroid distances. We average valid frames with at least two features and nonzero source-distance norm; the saved record also reports the nontrivial $n \geq 3$ subset. TW uses the standard rank penalty [9], with $k = 1$ shared by all

methods and datasets and the validity condition $k < n/2$. Larger $k = 3, 5$ settings are retained only on eligible frames; for example, ERA5 has just two valid $k = 3$ frames. A two-feature frame can fit its single distance exactly after scaling but cannot supply a valid $k = 1$ TW score.

TD sums matched-anchor absolute displacements. To compare display coordinates, each method's full-sequence canvas is mapped once to $[0, 1]$ using its actual raster extent; this is not per-frame normalization or occupied-range rescaling. TD is reported in canvas-width units and must be compared within a dataset. Stable leaf-row order resolves distance ties; the saved sensitivity uses the same tie-free frame subset for all methods.

Table S14: Standard geometry and stability re-evaluation on the unchanged main configurations.

Data	Method	SNS↓	TW_1 ↑	TD↓	N_{SNS}	N_{TW}
ERA5	TMTM	0.0799	0.8243	10.1108	115	88
ERA5	ST-MTM	0.0297	0.9653	7.1607	115	88
ERA5	RG-MTM-P	0.1186	0.7771	27.3609	115	88
ERA5	RG-MTM-R	0.1097	0.7554	23.8891	115	88
Fire	TMTM	0.2210	0.8170	11.2034	30	30
Fire	ST-MTM	0.0786	0.9195	7.9585	30	30
Fire	RG-MTM-P	0.0831	0.8793	8.0342	30	30
Fire	RG-MTM-R	0.0579	0.8725	3.2487	30	30
OLR	TMTM	0.1419	0.8135	8.7479	60	52
OLR	ST-MTM	0.0062	0.9751	2.3304	60	52
OLR	RG-MTM-P	0.0121	0.9587	8.9011	60	52
OLR	RG-MTM-R	0.0093	0.9634	8.5023	60	52
PM2.5	TMTM	0.0386	0.8021	2.6858	25	12
PM2.5	ST-MTM	0.0148	0.9896	1.0620	25	12
PM2.5	RG-MTM-P	0.1443	0.6194	6.2830	25	12
PM2.5	RG-MTM-R	0.1137	0.5986	6.1880	25	12
Ring	TMTM	0.1706	0.7067	1.3436	32	27
Ring	ST-MTM	0.0803	0.9172	2.3641	32	27
Ring	RG-MTM-P	0.1637	0.7087	3.5977	32	27
Ring	RG-MTM-R	0.1423	0.7149	2.3512	32	27
Gaussians	TMTM	0.3172	0.5247	0.1260	64	64
Gaussians	ST-MTM	0.0845	0.7533	0.2287	64	64
Gaussians	RG-MTM-P	0.1222	0.7533	0.3705	64	64
Gaussians	RG-MTM-R	0.0638	0.8639	0.5396	64	64

SNS and TW use support centroids. TD is summed in full-canvas-width units, not occupied-range-normalized mean displacement. N columns count valid frames; TW uses $k = 1$. All values are descriptive.

Relative to ST-MTM, RG-MTM-R has lower SNS on fire (0.0579 versus 0.0786) and Gaussians (0.0638 versus 0.0845), higher TW only on Gaussians, and lower TD on fire and Ring. The Ring TD difference is approximately 0.55%. ERA5, OLR and PM2.5 are worse on all three measures. RG-MTM-P has higher SNS and TD on every sequence. These results preclude a claim of universal geometry/stability improvement.

Standard rank correlations. Kendall τ_b and Spearman ρ compare the displayed anchors with the existing fixed-direction extremum references. A single optional reflection is fixed from the reference-minimum/maximum anchors in the first informative frame common to all methods (Ring: frame 8; others: frame 0); no per-frame reflection or whole-sequence score maximization is used. Main correlations average common valid $n \geq 3$ frames; raw-orientation and $n \geq 2$ summaries are also saved. The reference itself is fitted offline from the sequence, so these measure construction under a specified reference, not held-out prediction. High correlation does not imply low absolute displacement.

Motion classification. Using the same two-step and 2% thresholds as the main proxy, all tracked-pair windows receive approach, unchanged or separation labels. Balanced accuracy averages recalls of the true classes; macro-F1 averages classwise F1 with the three labels fixed. Unlike the clear-event ME subset, this also penalizes spurious changes in unchanged source pairs. These are computational labels, not participant responses. PM2.5 supplies only

ten windows (seven clear events), and Gaussian truth has only the unchanged class; we do not use that single-class case to assess motion classification.

Table S15: Standard order correlations and three-class geometric motion scores.

Data	Method	$\tau_b \uparrow$	$\rho \uparrow$	BA (%)↑	F1 (%)↑	Windows
ERA5	TMTM	-0.0011	-0.0181	48.2	47.3	127
ERA5	ST-MTM	-0.0072	0.0033	55.8	54.6	127
ERA5	RG-MTM-P	0.7653	0.8268	57.5	56.2	127
ERA5	RG-MTM-R	0.8924	0.9282	63.1	62.9	127
Fire	TMTM	0.3023	0.3409	50.9	45.8	238
Fire	ST-MTM	0.2039	0.2660	53.0	46.6	238
Fire	RG-MTM-P	0.8433	0.9132	55.7	47.9	238
Fire	RG-MTM-R	0.9793	0.9926	68.3	66.3	238
OLR	TMTM	-0.1351	-0.1808	41.7	41.7	63
OLR	ST-MTM	0.0101	0.0078	71.4	68.5	63
OLR	RG-MTM-P	0.9683	0.9812	91.7	89.2	63
OLR	RG-MTM-R	1.0000	1.0000	96.4	93.4	63
PM2.5	TMTM	0.5611	0.6333	19.4	19.0	10
PM2.5	ST-MTM	0.4500	0.5083	38.9	32.4	10
PM2.5	RG-MTM-P	0.6611	0.7250	61.1	62.4	10
PM2.5	RG-MTM-R	0.9111	0.9250	69.4	71.1	10
Ring	TMTM	-0.1855	-0.2561	32.9	31.9	209
Ring	ST-MTM	0.2892	0.3458	52.1	49.9	209
Ring	RG-MTM-P	0.6780	0.7778	52.2	46.2	209
Ring	RG-MTM-R	0.9111	0.9317	70.4	63.5	209
Gaussians	TMTM	0.4740	0.5828	–	–	237
Gaussians	ST-MTM	0.5573	0.6828	–	–	237
Gaussians	RG-MTM-P	0.5573	0.6828	–	–	237
Gaussians	RG-MTM-R	0.7292	0.7969	–	–	237

Correlations use the common $n \geq 3$ frames and one fixed orientation calibration; BA and macro-F1 use all two-step tracked-pair windows, including unchanged truth. Gaussian has only unchanged labels and is not used to assess motion classification. PM2.5 has only ten windows. These are computational proxies, not participant responses.

RG-MTM-R has higher reference-order correlations on all six sequences. It also has higher three-class balanced accuracy and macro-F1 on the five evaluable sequences than either baseline. These descriptive differences support the declared reference and motion tasks, not a combined quality ranking or a significance claim. Adjacent windows overlap. The actual raster evaluation changes one Ring RG-MTM-R decision relative to the historical continuous-anchor proxy (15 rather than 16 errors among 124 clear events); other main clear-event counts and errors agree.

Verification and evidence. TW agrees with scikit-learn 1.7.2 in 285 random comparisons to machine precision. Ninety SNS checks against independent numerical scale minimization, scale and reflection invariance, identity matching with births and deaths, and static-layout TD checks pass. The geometry and stability summaries are reproduced from saved landmark arrays. Direct pair counting and confusion matrices reproduce the new rank and classification summaries. Input hashes and original scientific sources are unchanged. The separate result folder `results/public_metrics_20261016/` contains source hashes, landmark arrays, protocols and metric summaries; historical result files retain their original keys and values.

S3.3 Sensitivity and additional evidence

Direction, extraction, tracking, persistence, and layout choices affect the tested quantities. The grouped sensitivity records contain eight fixed-direction checks and 1%, 2%, and 5% hierarchy-cost triggers; three change thresholds and four motion windows; ST-MTM parameter variants; extraction-scale and crop variants; and the four-month PM2.5 audit. They are exploratory analyses and use no multiplicity correction. Historical second-period ERA5 entries in source aggregates are excluded from the main six-sequence snapshot.

Persistence and feature population. In the retained ERA5 study, a 1% scalar-range cutoff removes 27.9% of leaves and leaves 23.7% of frames conflicting; at 2% these become 34.1% and 18.6%; at 5%, 46.2% and 10.2%. This control changes the leaf set, whereas RG-MTM relaxation retains extrema. Extraction at 150 km is evaluated through certified lower bounds only. The fire's weak background has not been varied.

Scalability. The retained binary diagnosis benchmark takes about 3.6 seconds for 4096 leaves on 16384 pixels. The degree-controlled record varies root degree while fixing widths, references, and pixel count. Its exponential subset recurrence is distinct from full-layout order enumeration. These timings are properties of the retained benchmark environment, not measured construction times for large scalar fields.

Dendrogram transfer. A retained experiment orders 1024 active fire cells by a dendrogram of their temporal signals. Its one-pixel-row model yields a positional certificate of 67.3% of display height; optimal leaf ordering reaches 88.2% maximum displacement. Geographic order has 20.6% maximum error, reflecting packing rather than hierarchy. This transfers the positional model only; no scalar-barrier or participant-performance result follows. Source: `dendrogram_demo.json`.

Failure conditions. Infeasible total packing requires changing geometry before certification. Strict extrema require sufficiently separated raster anchors; the renderer rejects collapsed gaps. A non-unique principal motion direction requires an external reference. A small hierarchy cost does not establish good two-dimensional distance preservation, and improved order alignment does not necessarily improve motion-proxy error, as shown by OLR. Cases with zero clear events remain undefined.

S4 Reproducibility and Evidence Roles

The accompanying repository contains the manuscript, experiment orchestration, method implementation, tests, and grouped numerical records. The no-figure source package contains only the manuscript and supporting LaTeX files. Raw data are excluded and require the local prepared data or the acquisition steps in `data/README.md`. The independent runner invokes existing scientific entry points and records commands, runtime, exit status, source hashes, and result hashes. It does not change their algorithms or reported thresholds.

Main evaluation. `experiments/publication2027/run.py` defaults to the six active sequences and the five rerun numerical stages. Explicit selection of all ten available stages also recomputes replication, exact frontiers, direction robustness, persistence controls, and witnesses. Retained sensitivity records are not labelled new runs. Per-stage logs and completion manifests reside under `results/publication2027/runs/`.

Evidence packaging. `experiments/publication2027/build.py` checks source hashes for positional and native-merge evidence, consistent frame counts, unique motion rows, and hierarchy lower bounds. It exports unrounded comparisons, a claim ledger, generated LaTeX tables, and source/snapshot hashes. Mixed aggregate files are filtered to the six active sequences in the main-evidence snapshot. Sensitivity, formal validation, protocols, and dendrogram transfer have separate folders. The original records remain available for historical traceability.

Numerical checks. The existing test suite includes independent LP comparisons, grid lower-bound checks, projected-motion margins, native sample and barrier semantics, preprocessing contracts, and no-event handling. The retained implementation-validation run passed 25 tests; the separate standard-metric re-evaluation adds metric and reconstruction checks without method changes. The validation checks cover numerical optimization, rendering semantics, and metric reconstruction.

Publication configuration. The main source uses the unmodified IEEE VGTC conference class in review mode. The main manuscript contains prose and equations, with numerical tables and pseudocode collected in this supplement. Both documents omit illustrations and figure reservations; pagination is unconstrained during this content revision. Submission ID 0 is to be replaced with the PCS identifier; Codex assistance is disclosed in the acknowledgments.

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